Key Innovations

- **Safe State Management**
- **Intelligent Optimization** of Resource Consumption (Pause/Resume)

Improved App Safety

- Preventing interactions after a Ref or Notifier has been disposed
  - Previous versions: interaction with disposed provider → unpredictable behavior
- Access a `Ref` or `Notifier` after disposal now immediately throws a clear error
- **Key Practice**: Provides technical grounds for mandatory `ref.mounted` checks

Wrapping in ProviderException

- Easier error debugging
- Errors during Provider reading are now wrapped in `ProviderException`
- Metadata Included: Which provider and which dependency caused the error
- Enables more specific root cause analysis during `try-catch`

Intelligent Pausing (Paused Providers)

- **The Highlight of Riverpod 3 Lifecycle**
- Automatic pausing of listeners for invisible widgets (Tabs, Navigator stacks, etc.)
- If all listeners are paused, the **Provider itself enters a paused state**
- Stops unnecessary computation/streams → Saves Battery & CPU (Auto-resumes when visible)

Maintaining Subscriptions (Atomic Rebuilds)

- Previous subscription relationships are kept during a rebuild until the new value is confirmed
- **Ensuring UI Continuity**

Summary & Conclusion

- **Automating Safety Measures and Performance Optimization**
- **Paused Listeners** feature provides significant optimization in complex apps